

Boom Components

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Primary Boom Angle Sensor

The primary boom angle sensor is used to limit the angle of the primary boom relative to the angle of the secondary boom and gravity.

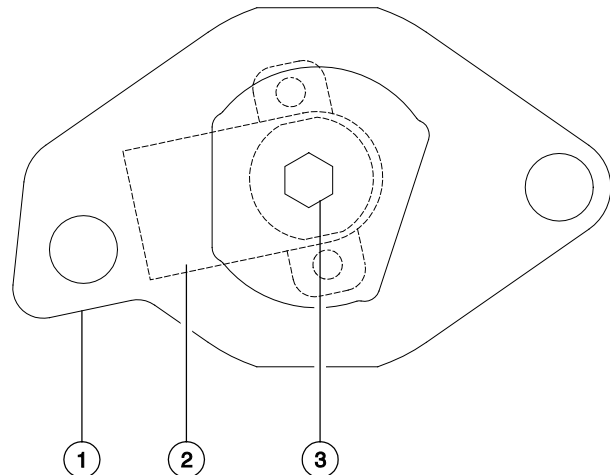
How to Replace the Primary Boom Angle Sensor

Note: Perform this procedure with the machine on a firm, level surface with the boom in the stowed position.

- 1 Remove the retaining fasteners from the boom end cover at the pivot end of the primary boom. Remove the cover from the machine.
- 2 Locate the primary boom angle sensor inside the primary boom at the boom pivot pin.
- 3 Disconnect the electrical connector from the sensor.
- 4 Remove the angle sensor mounting bracket retaining fasteners.
- 5 Remove the angle sensor and bracket assembly. Do not lose the two springs or the hex-shaped key.
- 6 Remove the sensor retaining fasteners from the mounting bracket. Remove the angle sensor.

Note: Note the sensor orientation on the bracket during removal so the new sensor will be installed in the same position.

- 7 Install the new angle sensor onto the mounting bracket.
- 8 Install one spring, removed in step 5, into the hex-shaped hole in the angle sensor. Install the other spring into the hex-shaped hole in the center of the boom pivot pin.
- 9 Insert the hex-shaped key into the angle sensor.
- 10 Align the side of the sensor with flat area of the boom pivot pin as shown below while holding the bracket away from key insertion.



- 1 flat area
- 2 angle sensor
- 3 hex-shaped hole

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- 11 While holding the hex-shaped key in position, rotate the angle sensor mounting bracket clockwise until it aligns with the machined pocket. Do not allow the key to rotate while holding it.
- 12 Insert the hex-shaped key into the boom pivot pin and the bracket into the machined pocket.

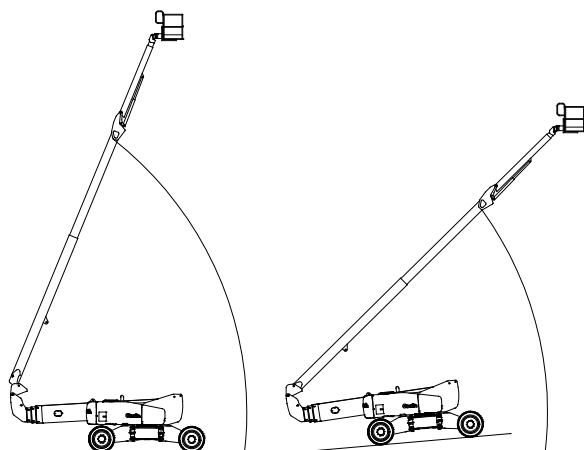
Note: Be sure the hex-shaped key remains engaged in the sensor.

- 13 Install the angle sensor bracket retaining fasteners.
- 14 Connect the electrical connector to the new sensor.
- 15 Install the primary boom end cover.
- 16 Calibrate the primary boom angle sensor. Refer to Repair Procedure, *How to Calibrate the Primary Boom Angle Sensor*.

How to Calibrate the Primary Boom Angle Sensor

The primary boom angle sensor is used to limit the angle of the primary boom relative to the angle of the secondary boom and gravity.

The primary boom maximum angle is gradually reduced as the downhill slope (positive degree Y axis) is increased. A flat level surface of 0° allows the primary boom maximum operational angle to reach 68°. On a 5° downhill slope the primary boom maximum operational angle is reduced to 43°.



0° slope, 68° boom angle

5° slope, 43° boom angle

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Note: If the axle sensor or secondary boom angle sensor have been removed or replaced, or the turntable level sensor has been replaced, they must be calibrated first. Refer to Repair Procedures, *How to Calibrate the Axle Angle Sensors* or *How to Calibrate the Secondary Boom Angle Sensor* or *How to Calibrate the Turntable Level Sensor*.

⚠ WARNING

Tip-over hazard. Failure to properly calibrate the machine could cause the machine to tip over resulting in death or serious injury.

Note: For software versions 4.01 and higher, use the 2 Point Calibration procedure. For software versions before 4.01, use the 6 Point Calibration procedure. The software version is displayed on the LCD screen when the red Emergency Stop button is pulled out to the on position.

Note: A digital level will only be required to perform the 6 Point Calibration procedure.

Note: A kit is available through Genie Product Support (Genie part number 58351). This kit includes a digital level with a magnetic base and cable harnesses.

Note: Use the following chart to identify the description of each LCD screen control button used in this procedure.



Plus



Minus



Previous



Enter

Note: Perform this procedure with the machine on a firm, level surface with the booms in the fully stowed position, in the drive enable zone and the axles fully extended.

6 Point Calibration procedure

Note: Use this procedure for software versions before 4.01.

Note: For the 2 Point Calibration procedure, proceed to step 37.

- 1 Push in the ground controls red Emergency Stop button to the off position. Do not turn the key switch to the off position.
- 2 Turn the key switch to ground controls.
- 3 Open the ground control box.

Before serial number 321:

- 4 Locate the 2 pin connector in the lower left hand area of the ground control box.
- 5 Remove the connector plug and insert a jumper wire between the two pins of the connector.

After serial number 320:

- 6 Locate the calibration toggle switch at the top of the ground control box. Activate calibration mode by moving the toggle switch in the left direction.
- 7 Temporarily install a control box door retaining fastener between the door and the box to prevent the control box door from moving the toggle switch while calibrating the machine.

All models:

- 8 Remove the key from the main key switch. Insert the key into the bypass/recovery key switch and turn it to the bypass position.

Note: The angle sensor calibration values will not be saved correctly unless the key switch is in the bypass position and the calibration toggle switch is activated.

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- 9 Press and hold the **enter** button on the ground control panel while pulling out the ground controls red Emergency Stop button to the on position. Hold the **enter** button for approximately 5 seconds and then release it.
- 10 Enter sensor calibration mode by pressing the buttons at the ground controls in the following sequence: **(plus)(enter)(enter)(plus)**.
- 11 Press the **enter** or **previous** button on the LCD screen until DELETE PRIMARY BOOM ANGLE SENSOR CALIBRATION is displayed.
- 12 Press the **plus** button to select YES, then press the **enter** button to accept.
- 13 Place a digital level that has been calibrated to gravity on top of the primary boom and note the angle displayed on the digital level.
- 14 At the PRIMARY BOOM ANGLE TO GRAVITY 0.0 DEG screen, press the **plus** button or **minus** button to adjust the display to the exact value shown on the digital level, then press the **enter** button.

Note: If the measured angle already matches the angle shown on the display at the ground controls, press the **plus** button or **minus** button to change the angle and then change back to the measured value. The system must detect a change in displayed value to record the calibrated value.
- 15 Start the engine from the ground controls.

Note: If the system exits out of calibration mode when the engine is started, repeat step 8 and then continue with step 14.
- 16 Fully raise the secondary boom. The boom is fully raised when the cylinder is fully extended and the boom stops moving.
- 17 Press and hold a function enable/speed select button and the primary boom down button and lower the primary boom down until the digital level displays -50 degrees.
- 18 At the ground controls, press the **plus** button or **minus** button to adjust the display to the exact value shown on the digital level and press the **enter** button.

Note: Be sure the number entered at the ground controls is a negative number.

Note: If the measured angle already matches the angle shown on the display at the ground controls, press the **plus** button or **minus** button to change the angle and then change back to the measured value. The system must detect a change in displayed value to record the calibrated value.
- 19 Press and hold a function enable/speed select button and the primary boom up button until the digital level displays -20 degrees.
- 20 At the ground controls, press the **plus** button or **minus** button to adjust the display to the exact value shown on the digital level and press the **enter** button.

Note: Be sure the number entered at the ground controls is a negative number.

Note: If the measured angle already matches the angle shown on the display at the ground controls, press the **plus** button or **minus** button to change the angle and then change back to the measured value. The system must detect a change in displayed value to record the calibrated value.
- 21 Press and hold a function enable/speed select button and the primary boom up button until the digital level displays 10 degrees.

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- 22 At the ground controls, press the **plus** button or **minus** button to adjust the display to the exact value shown on the digital level and press the **enter** button.

Note: If the measured angle already matches the angle shown on the display at the ground controls, press the **plus** button or **minus** button to change the angle and then change back to the measured value. The system must detect a change in displayed value to record the calibrated value.

- 23 Press and hold a function enable/speed select button and the primary boom up button until the digital level displays 40 degrees.

- 24 At the ground controls, press the **plus** button or **minus** button to adjust the display to the exact value shown on the digital level and press the **enter** button.

Note: If the measured angle already matches the angle shown on the display at the ground controls, press the **plus** button or **minus** button to change the angle and then change back to the measured value. The system must detect a change in displayed value to record the calibrated value.

- 25 Press and hold a function enable/speed select button and the primary boom up button until the primary boom is fully raised. The boom is fully raised when the cylinder is fully extended and the boom stops moving. The angle will be 70 degrees.

- 26 At the ground controls, press the **plus** button or **minus** button to adjust the display to the exact value shown on the digital level and press the **enter** button.

Note: If the measured angle already matches the angle shown on the display at the ground controls, press the **plus** button or **minus** button to change the angle and then change back to the measured value. The system must detect a change in displayed value to record the calibrated value.

- 27 Press and hold the engine start button for approximately 5 seconds to shut off the engine and to save the calibration settings.

Note: Do not turn the engine off with the key switch or red Emergency Stop button or all calibration points or values will not be saved.

- 28 Press the **enter** or **previous** button on the LCD screen until EXIT is displayed.
- 29 Press the **plus** button to select YES, then press the **enter** button to accept.
- 30 Turn the key back to the run position and remove the key from the bypass/recovery key switch. Insert the key into the main key switch and turn it to ground controls.

Note: Be sure that the bypass/recovery key switch is in the run position before attempting to operate the machine.

- 31 Wait approximately 20 seconds and turn the machine off by pressing the red Emergency Stop button in.

Before serial number 321

- 32 Remove the wire jumper from the 2 pin connector in the lower left hand area of the ground control box that was installed in step 5. Install the connector plug.

DANGER

Tip-over hazard. Failure to remove the jumper wire will result in an unsafe configuration which will compromise machine stability. Operating the machine in this configuration will result in the machine tipping over, resulting in death or serious injury. The jumper wire must be removed before operating the machine.

- 33 Close the ground control box and install the door retaining fasteners.

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After serial number 320:

- 34 Remove the fastener that was temporarily installed. Close the control box door and install the door retaining fasteners.

Note: When the control box door is closed, the calibration toggle switch is automatically activated to exit out of calibration mode.

All models:

- 35 Pull out the red Emergency Stop button. Start the engine and lower the boom to the stowed position. Be sure there are no calibration faults shown on the display.
- 36 Perform a primary boom angle test. Refer to Maintenance Procedure in the appropriate Service or Maintenance Manual, *Test the Primary Boom Angle Sensor*.

2 Point Calibration procedure

Note: Use this procedure for software versions 4.01 and higher.

- 37 Push in the ground controls red Emergency Stop button to the off position. Do not turn the key switch to the off position.
- 38 Turn the key switch to ground controls.
- 39 Open the ground control box.
- 40 Locate the calibration toggle switch at the top of the ground control box. Activate calibration mode by moving the toggle switch in the left direction.
- 41 Temporarily install a control box door retaining fastener between the door and the box to prevent the control box door from moving the toggle switch while calibrating the machine.
- 42 Remove the key from the main key switch. Insert the key into the bypass/recovery key switch and turn it to the bypass position.
- 43 Note: The angle sensor calibration values will not be saved correctly unless the key switch is in the bypass position and the calibration toggle switch is activated.
- 44 Press and hold the **enter** button on the ground control panel while pulling out the ground controls red Emergency Stop button to the on position. Hold the **enter** button for approximately 5 seconds and then release it.
- 45 Enter sensor calibration mode by pressing the buttons at the ground controls in the following sequence: **(plus)(enter)(enter)(plus)**.
- 46 Press the **enter** or **previous** button on the LCD screen until DELETE PRIMARY BOOM ANGLE SENSOR CALIBRATION is displayed.

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- 47 Press the **plus** button to select YES, then press the **enter** button to accept.
- 48 Lower the primary boom to the stowed position.
- 49 Press the **enter** or **previous** button on the LCD SCREEN UNTIL PRIMARY BOOM FULLY LOWERED is displayed.
- 50 Press the **plus** button to select YES, then press the **enter** button to accept.
- 51 Fully raise the secondary boom until it stops at the end of the cylinder stroke.
- 52 Fully raise the primary boom until it stops at the end of the cylinder stroke.
- 53 Press the **enter** or **previous** button on the LCD screen until PRIMARY BOOM FULLY RAISED is displayed.
- 54 Press the **plus** button to select YES, then press the **enter** button to accept.
- 55 Press and hold the engine start button for approximately 5 seconds to shut off the engine and to save the calibration settings.

Note: Do not turn the engine off with the key switch or red Emergency Stop button or all calibration points or values will not be saved.

- 56 Press the **enter** or **previous** button on the LCD screen until EXIT is displayed.
- 57 Press the **plus** button to select YES, then press the **enter** button to accept.

- 58 Turn the key back to the run position and remove the key from the bypass/recovery key switch. Insert the key into the main key switch and turn it to ground controls.

Note: Be sure that the bypass/recovery key switch is in the run position before attempting to operate the machine.

- 59 Wait approximately 20 seconds and turn the machine off by pressing the red Emergency Stop button in.
- 60 Remove the fastener that was temporarily installed. Close the control box door and install the door retaining fasteners.

Note: When the control box door is closed, the calibration toggle switch is automatically activated to exit out of calibration mode.

- 61 Pull out the red Emergency Stop button. Start the engine and lower the boom to the stowed position. Be sure there are no calibration faults shown on the display.
- 62 Perform a primary boom angle test. Refer to Maintenance Procedure in the appropriate Service or Maintenance Manual, *Test the Primary Boom Angle Sensor*.

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Secondary Boom Angle Sensor

The secondary boom angle sensor is used to limit the angle of the primary boom relative to the angle of the secondary boom and gravity.

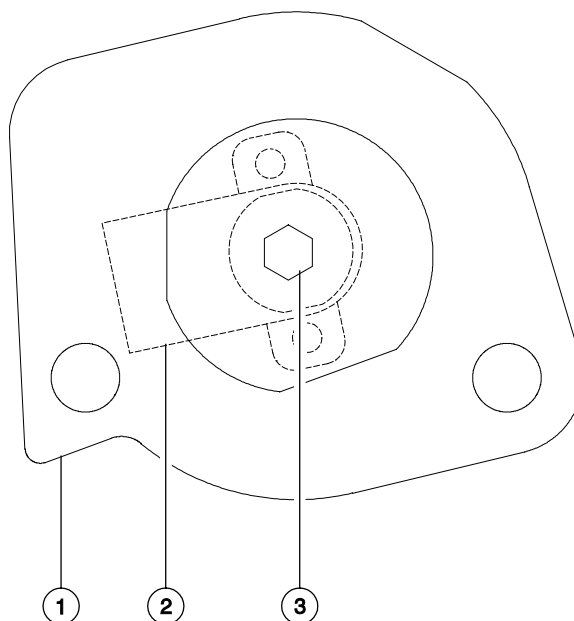
How to Replace the Secondary Boom Angle Sensor

Note: Perform this procedure with the machine on a firm, level surface with the boom in the stowed position.

- 1 Locate the secondary boom angle sensor on the turntable riser bulkhead at the ground controls side of the machine.
- 2 Disconnect the electrical connector from the sensor. Remove any tie wraps securing the wiring.
- 3 Remove the angle sensor cover retaining fastener. Remove the cover.
- 4 Remove the angle sensor mounting bracket retaining fasteners.
- 5 Remove the angle sensor and bracket assembly. Do not lose the two springs or the hex-shaped key.
- 6 Remove the sensor retaining fasteners from the mounting bracket. Remove the angle sensor.

Note: Note the sensor orientation on the bracket during removal so the new sensor will be installed in the same position.

- 7 Install the new angle sensor onto the mounting bracket.
- 8 Install one spring, removed in step 5, into the hex-shaped hole in the angle sensor. Install the other spring into the hex-shaped hole in the center of the boom pivot pin.
- 9 Insert the hex-shaped key into the angle sensor.
- 10 Align the side of the sensor with flat area of the boom pivot pin as shown below. Hold the bracket out to clear the side of the turntable side plate.



- 1 flat area
- 2 angle sensor
- 3 hex-shaped hole

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- 11 While holding the hex-shaped key in position, rotate the angle sensor mounting bracket clockwise until it aligns with the machined pocket. Do not allow the key to rotate while holding it.

Note: Be sure the hex-shaped key remains engaged in the sensor only.

- 12 Insert the hex-shaped key into the boom pivot pin and the bracket into the machined pocket.
- 13 Install the angle sensor bracket retaining fasteners.
- 14 Install the angle sensor cover and retaining fasteners.
- 15 Connect the electrical connector to the new sensor.
- 16 Calibrate the secondary boom angle sensor. Refer to Repair Procedure, *How to Calibrate the Secondary Boom Angle Sensor*.

How to Calibrate the Secondary Boom Angle Sensor

Note: If the axle sensor has been removed or replaced, or the turntable level sensor has been replaced, they must be calibrated first. Refer to Repair Procedure, *How to Calibrate the Axle Angle Sensors* or *How to Calibrate the Turntable Level Sensor*.

⚠ WARNING

Tip-over hazard. Failure to properly calibrate the machine could cause the machine to tip over resulting in death or serious injury.

Note: For software versions 4.01 and higher, use the 2 Point Calibration procedure. For software versions before 4.01, use the 6 Point Calibration Procedure. The software version is displayed on the LCD screen when the red Emergency Stop button is pulled out to the on position.

Note: A digital level will only be required to perform the 6 Point Calibration procedure.

Note: A kit is available through Genie Product Support (Genie part number 58351). This kit includes a digital level with a magnetic base and cable harnesses.

Note: Perform this procedure with the machine on a firm, level surface with the booms in the fully stowed position, in the drive enable zone and the axles fully extended.

Note: Use the following chart to identify the description of each LCD screen control button used in this procedure.



Plus



Minus



Previous



Enter

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6 Point Calibration procedure

Note: Use this procedure for software versions before 4.01.

Note: For the 2 Point Calibration procedure, proceed to step 36.

- 1 Push in the ground controls red Emergency Stop button to the off position. Do not turn the key switch to the off position.
- 2 Turn the key switch to ground controls.
- 3 Open the ground control box.

Before serial number 321:

- 4 Locate the 2 pin connector in the lower left hand area of the ground control box.
- 5 Remove the connector plug and insert a jumper wire between the two pins of the connector.

After serial number 320:

- 6 Locate the calibration toggle switch at the top of the ground control box. Activate calibration mode by moving the toggle switch in the left direction.
- 7 Temporarily install a control box door retaining fastener between the door and the box to prevent the control box door from moving the toggle switch while calibrating the machine.

All models:

- 8 Remove the key from the main key switch. Insert the key into the bypass/recovery key switch and turn it to the bypass position.

Note: The angle sensor calibration values will not be saved correctly unless the key switch is in the bypass position and the calibration toggle switch is activated.

- 9 Press and hold the **enter** button on the ground control panel while pulling out the ground controls red Emergency Stop button to the on position. Hold the **enter** button for approximately 5 seconds and then release it.
- 10 Enter sensor calibration mode by pressing the buttons at the ground controls in the following sequence: **(plus)(enter)(enter)(plus)**.

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- 11 Press the **enter** or **previous** button on the LCD screen until DELETE SECONDARY BOOM ANGLE SENSOR CALIBRATION is displayed.
- 12 Press the **plus** button to select YES, then press the **enter** button to accept.
- 13 Place a digital level that has been calibrated to the Y axis of the turntable on top of the secondary boom and note the angle displayed on the digital level.
- 14 At the SECONDARY BOOM ANGLE TO GRAVITY -3.5DEG screen, press the **plus** button or **minus** button to adjust the display to the exact value shown on the digital level and press the **enter** button.

Note: Be sure the number entered at the ground controls is a negative number.

Note: If the measured angle already matches the angle shown on the display at the ground controls, press the **plus** button or **minus** button to change the angle and then change back to the measured value. The system must detect a change in displayed value to record the calibrated value.

- 15 Start the engine from the ground controls.

Note: If the system exits out of calibration mode when the engine is started, repeat step 8 and then continue with step 14.

- 16 Press and hold a function enable/speed select button and the secondary boom up button until the digital level displays 20 degrees.

- 17 At the ground controls, press the **plus** button or **minus** button to adjust the display to the exact value shown on the digital level and press the **enter** button.

Note: If the measured angle already matches the angle shown on the display at the ground controls, press the **plus** button or **minus** button to change the angle and then change back to the measured value. The system must detect a change in displayed value to record the calibrated value.

- 18 Press and hold a function enable/speed select button and the secondary boom up button until the digital level displays 35 degrees.
- 19 At the ground controls, press the **plus** button or **minus** button to adjust the display to the exact value shown on the digital level and press the **enter** button.

Note: If the measured angle already matches the angle shown on the display at the ground controls, press the **plus** button or **minus** button to change the angle and then change back to the measured value. The system must detect a change in displayed value to record the calibrated value.

- 20 Press and hold a function enable/speed select button and the secondary boom up button until the digital level displays 50 degrees.
- 21 At the ground controls, press the **plus** button or **minus** button to adjust the display to the exact value shown on the digital level and press the **enter** button.

Note: If the measured angle already matches the angle shown on the display at the ground controls, press the **plus** button or **minus** button to change the angle and then change back to the measured value. The system must detect a change in displayed value to record the calibrated value.

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22 Press and hold a function enable/speed select button and the secondary boom up button until the digital level displays 65 degrees.

23 At the ground controls, press the **plus** button or **minus** button to adjust the display to the exact value shown on the digital level and press the **enter** button.

Note: If the measured angle already matches the angle shown on the display at the ground controls, press the **plus** button or **minus** button to change the angle and then change back to the measured value. The system must detect a change in displayed value to record the calibrated value.

24 Press and hold a function enable/speed select button and the secondary boom up button. Fully raise the secondary boom. The boom is fully raised when the cylinder is fully extended and the boom stops moving. The angle will be 76 degrees.

25 At the ground controls, press the **plus** button or **minus** button to adjust the display to the exact value shown on the digital level and press the **enter** button.

Note: If the measured angle already matches the angle shown on the display at the ground controls, press the **plus** button or **minus** button to change the angle and then change back to the measured value. The system must detect a change in displayed value to record the calibrated value.

26 Press and hold the engine start button for approximately 5 seconds to shut off the engine and to save the calibration settings.

Note: Do not turn the engine off with the key switch or red Emergency Stop button or all calibration points or values will not be saved.

27 Press the **enter** or **previous** button on the LCD screen until EXIT is displayed.

28 Press the **plus** button to select YES, then press the **enter** button to accept.

29 Turn the key back to the run position and remove the key from the bypass/recovery key switch. Insert the key into the main key switch and turn it to ground controls.

Note: Be sure that the bypass/recovery key switch is in the run position before attempting to operate the machine.

30 Wait approximately 20 seconds and turn the machine off by pressing the red Emergency Stop button in.

Before serial number 321:

31 Remove the wire jumper from the 2 pin connector in the lower left hand area of the ground control box that was installed in step 5. Install the connector plug.

DANGER

Tip-over hazard. Failure to remove the jumper wire will result in an unsafe configuration which will compromise machine stability. Operating the machine in this configuration will result in the machine tipping over, resulting in death or serious injury. The jumper wire must be removed before operating the machine.

32 Close the ground control box and install the door retaining fasteners.

Boom Components

After serial number 320:

- 33 Remove the fastener that was temporarily installed. Close the control box door and install the door retaining fasteners.

Note: When the control box door is closed, the calibration toggle switch is automatically activated to exit out of calibration mode.

All models:

- 34 Pull out the red Emergency Stop button. Start the engine and lower the boom to the stowed position. Be sure there are no calibration faults shown on the display.
- 35 Perform secondary boom angle test. Refer to Maintenance Procedure in the appropriate Service or Maintenance Manual, *Test the Secondary Boom Angle Sensor*.

2 Point Calibration procedure

Note: Use this procedure for software versions 4.01 and higher.

- 36 Push in the ground controls red Emergency Stop button to the off position. Do not turn the key switch to the off position.
- 37 Turn the key switch to ground controls.
- 38 Open the ground control box.
- 39 Locate the calibration toggle switch at the top of the ground control box. Activate calibration mode by moving the toggle switch in the left direction.
- 40 Temporarily install a control box door retaining fastener between the door and the box to prevent the control box door from moving the toggle switch while calibrating the machine.
- 41 Remove the key from the main key switch. Insert the key into the bypass/recovery key switch and turn it to the bypass position.

Note: The angle sensor calibration values will not be saved correctly unless the key switch is in the bypass position and the calibration toggle switch is activated.

- 42 Press and hold the **enter** button on the ground control panel while pulling out the ground controls red Emergency Stop button to the on position. Hold the **enter** button for approximately 5 seconds and then release it. Enter sensor calibration mode by pressing the buttons at the ground controls in the following sequence:
(plus)(enter)(enter)(plus).

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- 43 Press the **enter** or **previous** button on the LCD screen until DELETE SECONDARY BOOM ANGLE SENSOR CALIBRATION is displayed.
- 44 Press the **plus** button to select YES, then press the **enter** button to accept.
- 45 Lower the secondary boom to the stowed position.
- 46 Press the **enter** or **previous** button on the LCD SCREEN UNTIL SECONDARY BOOM FULLY LOWERED is displayed.
- 47 Press the **plus** button to select YES, then press the **enter** button to accept.
- 48 Fully raise the secondary boom until it stops at the end of the cylinder stroke.
- 49 Press the **enter** or **previous** button on the LCD SCREEN UNTIL SECONDARY BOOM FULLY RAISED is displayed.
- 50 Press the **plus** button to select YES, then press the **enter** button to accept.
- 51 Press and hold the engine start button for approximately 5 seconds to shut off the engine and to save the calibration settings.

Note: Do not turn the engine off with the key switch or red Emergency Stop button or all calibration points or values will not be saved.

- 52 Press the **enter** or **previous** button on the LCD screen until EXIT is displayed.

- 53 Press the **plus** button to select YES, then press the **enter** button to accept.
- 54 Turn the key back to the run position and remove the key from the bypass/recovery key switch. Insert the key into the main key switch and turn it to ground controls.

Note: Be sure that the bypass/recovery key switch is in the run position before attempting to operate the machine.

- 55 Wait approximately 20 seconds and turn the machine off by pressing the red Emergency Stop button in.
- 56 Remove the fastener that was temporarily installed. Close the control box door and install the door retaining fasteners.

Note: When the control box door is closed, the calibration toggle switch is automatically activated to exit out of calibration mode.

- 57 Pull out the red Emergency Stop button. Start the engine and lower the boom to the stowed position. Be sure there are no calibration faults shown on the display.
- 58 Perform secondary boom angle test. Refer to Maintenance Procedure in the appropriate Service or Maintenance Manual, *Test the Secondary Boom Angle Sensor*.

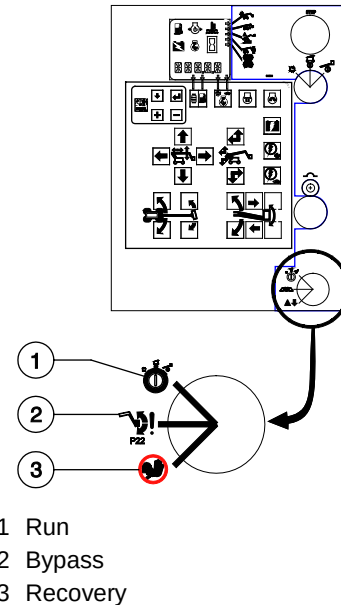
Ground Controls

6-1 Bypass/Recovery Key Switch

The turntable control box (TCON) is the communication and operations center for the machine. The turntable control box contains two key switches. The main key switch towards the top of the control box is for selection of ground or platform controls. The key switch at the bottom of the control box is the Bypass/Recovery key switch. Bypass and Recovery modes are only intended for certain circumstances and are not part of normal machine operation. If either the Bypass or the Recovery function is required, this indicates there may be faults with the machine. Contact trained personnel immediately.

Bypass is used for a platform out-of-level condition and calibrating certain machine parameters.

Recovery is only to be used as a last attempt to lower the platform when the operator in the platform is unable to do so, system failure or in emergency situations.



Note: The bypass/recovery key switch is located on the lower right side of the control box from serial number Z13512-1712.

Ground Controls

How to Use the Bypass Mode

⚠ DANGER

Tip-over hazard. Operating the machine outside of the operating envelope while in Bypass mode will result in death or serious injury if proper operating procedures and safety precautions are not followed. Do not use this mode if you are not trained and familiar with the operating envelope of the machine.

Note: Before using the Bypass mode, make sure you understand the fault code or issue affecting the operation of the machine to be sure the use of bypass is required.

The Bypass mode will allow the platform to be manually leveled when an out-of-level condition exists. In the event that the platform angle is greater than 10° from level, the boom angle and platform level functions are disabled. Use of the Bypass mode will allow the platform to be manually adjusted to within the normal operating envelope, $\pm 4.5^\circ$. Only auxiliary power can be used to correct an out of level platform fault.

- 1 Turn the engine off.
- 2 Turn the main key switch to ground controls. Remove the key from the main key switch and insert the key into the bypass/recovery key switch.

Note: The main key switch must remain in the ground control position.

- 3 Turn the bypass/recovery key switch to the bypass position.

- 4 Using auxiliary power, operate the ground control buttons to level the platform.

Note: Only the auxiliary power unit can be used to correct an out of level platform fault.

- 5 Turn the bypass/recovery key switch to the run position.
- 6 Remove the key from the bypass/recovery key switch and insert the key into the main key switch.

Note: If the Bypass function has been used, there may be faults with the machine. Check the LCD screen on the ground control box for machine faults, then contact trained service personnel.

Ground Controls

How to Use the Recovery Mode

Recovery is only to be used as a last attempt to lower the platform when the operator in the platform is unable to do so, system failure or in emergency situations.

⚠ WARNING

Bodily injury hazard. When using recovery mode, the platform may not fully lower to the ground when the recovery mode is completed. Failure to use only suitable equipment and/or practices to allow the operator to safely exit the platform could result in death or serious injury.

⚠ WARNING

Bodily injury hazard. Platform leveling is not active when using recovery mode. The platform could reach high out-of-level conditions when using this mode. The operator will need to secure themselves to the platform to prevent falling injury.

The Recovery mode allows the platform to be lowered in the event the operator in the platform is unable to lower the platform using the platform controls, system failure or emergency situations. The recovery sequence will automatically retract the primary boom, retract the secondary boom and then lower the primary boom using the auxiliary power unit to allow the operator at the platform controls to exit the platform.

Note: This procedure is for machines using software version 1.11 and later. For machines using software version 1.01, contact Genie Product Support.

- 1 Turn the main key switch to ground controls. Remove the key from the main key switch and insert the key into the bypass/recovery key switch.

Note: If this procedure is performed with the main key switch in the off position an active latched safety fault will be set and will have to be cleared.

- 2 Turn and hold the bypass/recovery key switch to the recovery position. The switch must be held in the recovery position.
- ⦿ Result: The auxiliary power unit will turn on and the boom will begin the following recovery sequence.
 - The primary boom will retract.
 - The secondary boom will retract.
 - The primary boom will lower.

Note: The key switch must be held in the recovery position until the recovery sequence is complete or until the operator in the platform can safely exit the platform.

Note: If any boom safety limit switches are faulty, the boom will only retract and not lower and the operator will need to be recovered from that point.

Note: If the event the platform becomes out of level $\pm 15^\circ$, all functions using auxiliary power will be disabled, PLATFORM LEVEL > 15 DEGREES will be displayed on the ground control LCD screen, and the operator will have to be recovered using the Recovery Mode.

Note: If the Recovery function has been used, this may indicate there may be faults with the machine. Tag and remove the machine from service until the fault has been corrected by trained personnel.

Ground Controls

6-4 Full Machine Calibration

Full machine calibration must be completed in the proper sequence when the ALC-1000 circuit board (TCON) in the ground control box has been replaced or the turntable level sensor (SCON) has been replaced.

How to Fully Calibrate the Machine

Calibration procedures shall only be completed by qualified technicians that have Genie factory service training.

⚠ WARNING

Tip-over hazard. Failure to calibrate the machine in the proper sequence could cause the machine to tip over resulting in death or serious injury.

Note: A digital level will be required to perform this procedure.

Note: A kit is available through Genie Product Support (Genie part number 58351). This kit includes a digital level with a magnetic base and cable harnesses.

Note: Use the following chart to identify the description of each LCD screen control button used in this procedure.



Plus



Minus



Previous



Enter

Note: Start this procedure with the booms in the fully stowed position and the axle retracted.

Full machine calibration must be completed in the following sequence:

- Select engine configuration. Refer to Repair Section, Display Module. *Unit of Measure and Language*.
- Joysticks. Refer to Repair Procedure, *How to Calibrate a Joystick*.
- Turntable level sensor. Refer to Repair Procedure, *How to Calibrate the Turntable Level Sensor*.
- Platform level sensor. Refer to Repair Procedure, *How to Calibrate the Platform Level Sensor*.
- Axle angle sensors. Refer to Repair Procedure, *How to Calibrate the Axle Angle Sensors*.
- Steer sensors. Refer to Repair Procedure, *How to Calibrate All Steer Sensors*.
- Secondary boom angle sensor. Refer to Repair Procedure, *How to Calibrate the Secondary Boom Angle Sensor*.
- Primary boom angle sensor. Refer to Repair Procedure, *How to Calibrate the Primary Boom Angle Sensor*.
- Jib boom bellcrank angle sensor. Refer to Repair Procedure, *How to Calibrate the Jib Boom Bellcrank Sensor*.
- Select option configuration. Refer to Repair Section, Display Module. *Options*.

Display Module

This table lists the various screens and menu options of the operating software. Some display menus are for informational purpose only, while others can be used to change the machine operating parameters.

⚠ WARNING Tip-over hazard. Calibration and parameter settings must be completed by a person trained and qualified on the repair of this machine. Failure to properly calibrate or set parameters could cause the machine to tip over resulting in death or serious injury.

Note: The key switch must be in the off position before entering the programming mode.

Note: Use the following chart to identify the description of each LCD screen control button used in this procedure.



Plus



Minus



Previous



Enter

Activation of the **enter** or **previous** buttons scrolls through the screens. To change parameter values or select a setting, use the **plus** button (to increase or scroll forward) and the **minus** button (to decrease or scroll backwards). Then press the **enter** button to save the new value to memory. An audible beep will indicate a save to memory. Use the **enter** and **previous** buttons to scroll to the EXIT menu. Use the **plus** button to change to YES and use the **enter** button to exit.

Screen or Menu	Procedure	Description	Range or Selection
Operator	Default	Hourmeter (on power up) Engine speed Engine oil pressure PSI (English) Engine oil pressure kPa (metric) Engine temperature °F (English) Engine temperature °C (metric) Primary boom angle to gravity Turntable level sensor X° direction Turntable level sensor Y° direction Platform level sensor degree Battery volts	engine temp will not display until temp is >100°F engine temp will not display until temp is >38°C

Display Module

Screen or Menu	Procedure	Description	Range or Selection
Machine Status	With key switch on, press the (plus)(minus) buttons at the same time.	Hydraulic pressure PSI (English) Hydraulic pressure kPa (metric) Primary boom to secondary boom angle Primary boom length Secondary boom angle* Secondary boom length Jib bellcrank angle DPF Regeneration Mode (Auto / Force / Inhibit)	0-4500 PSI 0-31000 kPa +22° to +136° =0', >0', >22 -3.5° to 76° =0FT, >0 FT -10° to +10° *(referenced to chassis tilt angle)
Unit of Measure and Language	With key switch OFF, press and hold the enter button and turn key switch to on position. Release the enter button and press (plus)(minus)(minus)(plus) . Overload Recovery (software V3.07 and later)	Metric/English (measurement units) Set engine Clear Overload Recovery	English, German, French, Spanish, Portuguese, Italian, Dutch, and Swedish. Deutz TD2011L04i (DL04i) Perkins 1104D-44T (P1104) YES/NO A passcode is required to clear the message
Default Reset Software version 3.11 or lower and software version 4.01	With key switch OFF, press and hold the enter button and turn key switch to on position. Release the enter button and press (minus)(minus)(previous)(previous)	Delete drive functions Delete boom function speed Delete lift function ramps Delete all (Contact Genie Product Support before using this option) Delete Faults	Delete faults will reset active latching faults. Delete faults will not clear fault history.